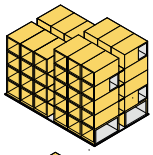
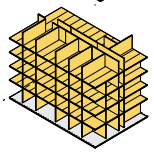
4.3 MID-RISE RESIDENTIAL

Due to the demand for housing in urban areas, mid-rise residential buildings account for a large proportion of development in the UK. The structural properties of mass timber systems mean they are particularly suitable for buildings of 5-10 storeys, and the prefabrication of mass timber components enables buildings on constrained sites to be more easily constructed, more quickly.

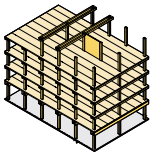
Suitable systems



Volumetric
Utilises the highest level of prefabrication.



Panellised
Fewer follow on trades as more walls are prefabricated.



Post and Beam
Improved level of internal layout flexibility.

Building height considerations

Mass timber systems have been used to construct residential buildings of up to ten storeys in the UK, and taller buildings internationally, demonstrating their viability and capability.

In England, building regulations prejudice against using timber in the external walls of residential buildings above 18m.

A solution for developments above this threshold can be hybrid systems where an internal mass timber structure is combined with a non-combustible façade (e.g. light steel frame) and structural core of mass timber or, more commonly, in-situ concrete.

Structural grid considerations

The repetitive and cellular layout of residential buildings lends itself to the use of panellised and volumetric systems; as vertical load paths are consistent. It is important to work to system parameters to maximise efficiency and minimise costs.

For panellised, a 6m grid is recommended and works well with most apartment typologies. The grid for volumetric modular will be influenced by transport restrictions on delivery to site.

Post and beam, and hybrid systems, are typically the most flexible as spans can be larger, although a 6m span in one direction is desirable to enable the efficient use of CLT slabs.

Fire safety considerations

Timber buildings are subject to the same statutory regulations as traditionally constructed buildings, with the requirements for external walls in tall residential buildings being a key consideration.

The risk of fire in a timber building is highest during construction, so it is important that a fire risk assessment and robust site management plan is in place to protect the timber system at all times.